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11 June 2026

Editor-in-Chief
The American Mathematical Monthly
Mathematical Association of America

Dear Editor,

I submit for your consideration the manuscript:

**The Continuous-Coefficient Jensen Equation:
How Real Mixing Weights Retire Regularity**

for publication in *The American Mathematical Monthly* as an **Article**.

What the paper does. The paper shows that the continuous-coefficient Jensen equation $G(pu_1 + (1 - p)u_2) = pG(u_1) + (1 - p)G(u_2)$ for all $u_1, u_2 \in [0, M]$ and all $p \in [0, 1]$ forces G to be affine with *no regularity hypothesis*: a one-line endpoint substitution ($u_1 := M$, $u_2 := 0$, $p := v/M$) reads $G(v)$ directly off the straight line through the endpoints. The paper accompanies this one-line observation with (i) a dictionary comparing the result with the well-known case of the discrete-coefficient Jensen equation (where Hamel’s pathology is real and regularity *is* needed), (ii) a structural explanation of why the two cases differ, and (iii) a diagnostic that separates the saturated-Jensen settings in which these regularity hypotheses are vestigial from the more common settings in which they remain load-bearing. The same five hypotheses appear on both sides of the line: in the axiomatic characterization of Shannon entropy they are indispensable; in a calibration computation on an atomless probability space they are not. A worked quantum example — mixture-affine assignments on density operators — shows the asymmetry alive in a single modern literature: Gleason-type derivations of the Born rule spend exactly the dictionary’s hypotheses on the measurement side, while the preparation side, by the convex-domain form of the theorem, spends none.

Why the Monthly. The core result is elementary enough for a second-year analysis student to verify in an afternoon, and yet modern applied work spans both sides of a sharp line: in some saturated-Jensen settings the continuous-coefficient equation makes the classical regularity hypotheses vestigial, while in others — superficially identical — those hypotheses remain indispensable. The article supplies the diagnostic that tells them apart, and (in its closing note on provenance) recounts how the author was led to it by an instance of the rarer error of adding a hypothesis that was not needed. The Monthly is the natural home for an expository article that makes this asymmetry explicit, mirrors the regularity dictionary, and invites readers to audit their own derivations.

Novelty and relationship to prior art. The substitution technique that closes Theorem 1 is folklore: Aczél’s 1966 monograph (§2.1.4) contains a dyadic precursor proved under a continuity hypothesis, and the genre is what Kuczma’s 2009 introduction describes as the part of Jensen-equation theory that does not require regularity assumptions on the function. The precise continuous-coefficient regularity-free statement of Theorem 1 does not, however, appear in this exact form in the standard references the author consulted. The contribution of this article is therefore not Theorem 1 itself but its *explicit packaging*: the dictionary contrasting the two Jensen forms, the structural explanation of the Hamel-pathology mechanism, and the identification of a recurrence pattern in derivations that produce the continuous-coefficient form as a saturated identity. A closing note records that the observation reached the author through a formalization project on a separate topic—a route that may interest readers curious about how mechanized proof can pressure-test classical heuristics.

Submission details.

- **Manuscript type:** Article (expository).
- **Page count:** approximately 11 pages anonymous body, 12 pages with title page.
- **Word count (approximate):** 4,200 words (TeX-counted body).
- **MSC 2020 primary:** 39B22 (Functional equations for real functions).
- **MSC 2020 secondary:** 26B25 (Convexity), 91B16 (Utility theory), 94A17 (Measures of information; entropy).
- **Double-blind:** Yes. The anonymous body (`manuscript-anon.pdf`) contains no author-identifying information; the file with author details (`manuscript.pdf`) carries the title page on page 1.
- **Preprint status:** The manuscript has not been deposited on arXiv. It is not under consideration elsewhere.
- **Competing interests:** None.
- **Funding:** This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.
- **Use of generative AI:** Disclosed in the manuscript’s Acknowledgments section. A proprietary neuro-symbolic AI research harness employing Claude Opus 3.7 and Lean 4 was used during initial exploratory theorem-discovery, and Claude Fable 5 was used for editorial review and formatting assistance. All results and presentation were analyzed and carried out by the author, who assumes full responsibility.

I look forward to the referees’ comments.

Sincerely,

Ali Elouafiq